EC39001: Analog Communication Lab Experiment 4: Envelope Detector and Amplitude Modulation using Log-Antilog Circuit

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1 Aim of the Experiment

The objectives of this experiment are :

1. To design and implement an **envelope detector** to demodulate an amplitude-modulated (AM) signal, thereby recovering the original message signal. This involves amplifying the AM signal and using a buffered RC circuit followed by a low-pass filter.
2. To design and demonstrate an alternative method of **amplitude modulation** using a **log-antilog circuit**. This involves using the property that the sum of logarithms corresponds to the logarithm of a product.

2 Components Used

- Breadboard
- LM741 Operational Amplifiers
- 1N4007 Silicon Diodes
- Potentiometer: $5k\Omega$
- Function Generator
- Mixed Signal Oscilloscope
- Resistors: 100Ω (2), $1k\Omega$, $1.5k\Omega$, $3.3k\Omega$, $4.7k\Omega$, $10k\Omega$, $18k\Omega$, $47k\Omega$, $100k\Omega$

- Capacitors: $5.6nF$, $0.1\mu F$, $470nF$, $22\mu F$
- $\pm 12V$ DC Power Supply
- Connecting Wires

3 Theory

3.1 Envelope Detector

An envelope detector is a simple and effective circuit for demodulating an AM signal. The core idea is to have a circuit that follows the peaks, or the "envelope," of the modulated carrier wave. This is achieved with a diode and an RC circuit.

The diode rectifies the incoming AM signal, allowing only the positive half-cycles to pass. The capacitor charges up quickly to the peak voltage of these cycles. When the input voltage drops below the capacitor's voltage, the diode becomes reverse-biased, and the capacitor begins to discharge slowly through the resistor.

The time constant, $\tau = RC$, is critical. It must be chosen carefully to satisfy the condition:

$$\frac{1}{f_c} \ll \tau \ll \frac{1}{f_m}$$

where f_c is the carrier frequency and f_m is the message frequency.

- If τ is too small, the capacitor discharges too quickly between carrier peaks, resulting in a large ripple in the output.
- If τ is too large, the capacitor cannot discharge fast enough to follow the envelope when the message signal's amplitude is decreasing, a phenomenon known as **diagonal clipping**.

The output from the RC circuit is a DC-shifted version of the message signal with some ripple. This is then passed through a buffer (to prevent loading) and a low-pass filter to remove the remaining carrier frequency components and block the DC offset, thus recovering the original message signal.

3.2 Log-Antilog Amplitude Modulator

This method of modulation leverages the fundamental property of logarithms:

$$\log(A) + \log(B) = \log(A \times B)$$

$$\exp(\log(A \times B) = A \times B$$

By converting two signals into their logarithmic equivalents, adding them, and then taking the anti-logarithm (exponentiation), we can effectively multiply them. An AM signal is essentially the product of the message and carrier signals (with appropriate DC offsets).

The circuit uses operational amplifiers (op-amps) with diodes in the feedback path. The exponential current-voltage $(I - V)$ relationship of a diode is exploited to create these logarithmic and anti-logarithmic responses.

- **Log Amplifier:** An op-amp with a diode in its feedback loop produces an output voltage proportional to the logarithm of the input voltage.

- **Summing Amplifier:** A standard op-amp adder circuit is used to sum the logarithmic outputs.
- **Anti-Log Amplifier:** An op-amp with a diode at its input produces an output voltage proportional to the exponent of the input voltage.

Crucially, since the logarithm is only defined for positive values, both the message signal $m(t)$ and the carrier signal $c(t)$ must be given appropriate DC offsets to ensure their instantaneous values are always greater than zero.

4 Circuit Diagrams

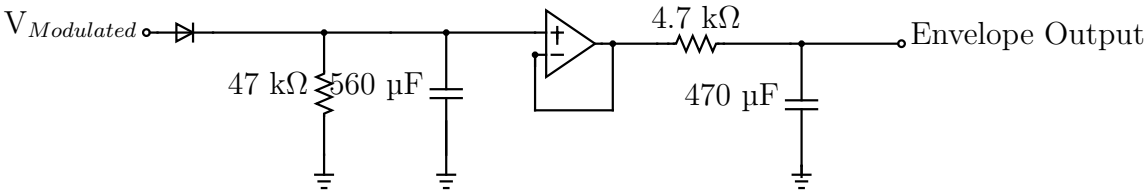


Figure 1: Circuit Diagram for the Envelope Detector with Buffer and Low-Pass Filter

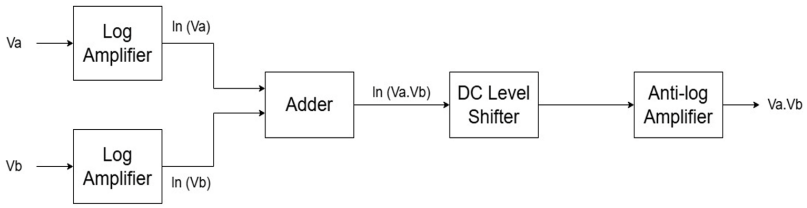


Figure 2: Circuit Diagram for the complete Log-Antilog Amplitude Modulator

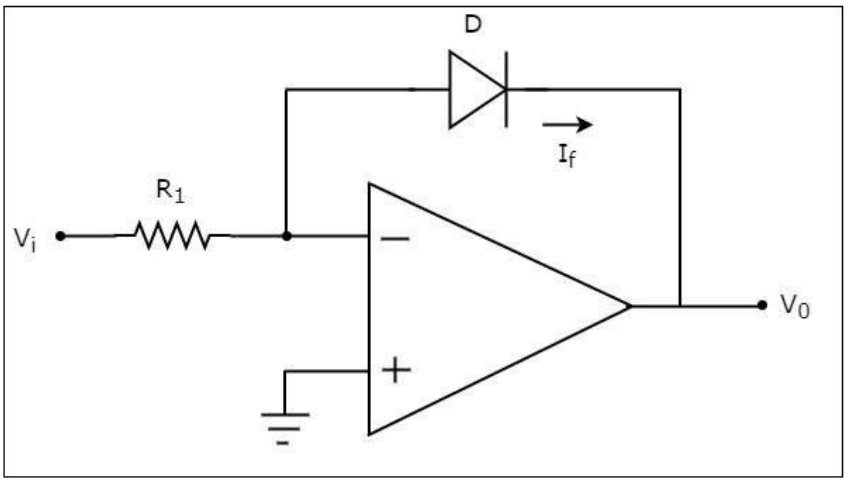


Figure 3: Circuit Diagram for Log Circuit

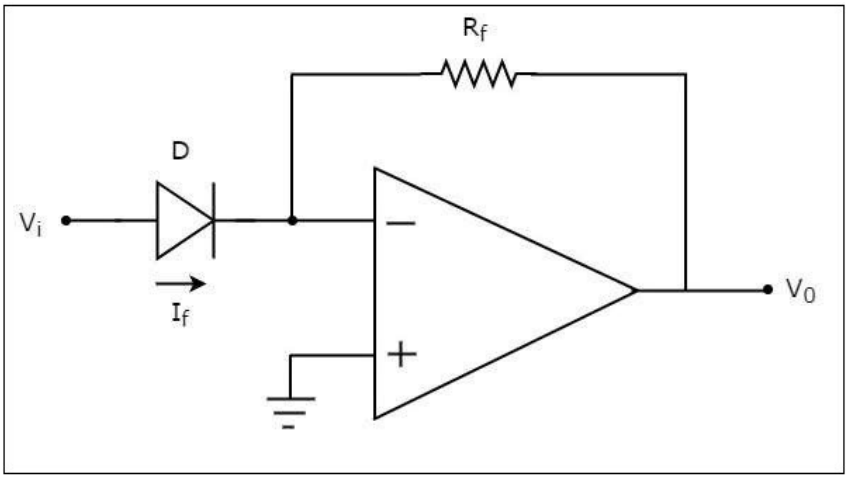


Figure 4: Circuit Diagram for AntiLog Circuit

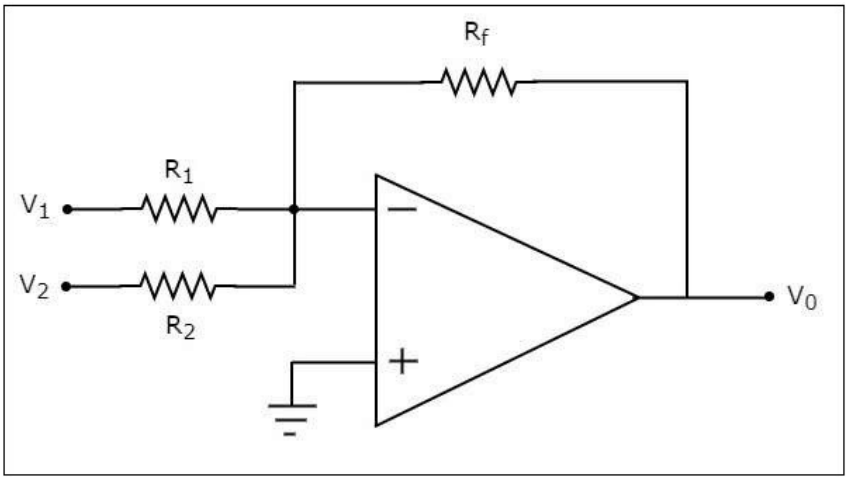


Figure 5: Circuit Diagram for Adder Circuit

5 Observations and Results

5.1 Envelope Detector Performance

The under-modulated AM signal was fed into the envelope detector circuit. The carrier frequency was $20\,kHz$ and the message frequency was $2\,kHz$. The output of the detector, after passing through the low-pass filter (cutoff $\approx 3\,kHz$), successfully recovered the message signal.



Figure 6: Input AM signal (Blue) and Rectified Message Signal (Yellow) with its FFT

In figure 6 we can see rectified message signal envelope with the some high frequency component introduced.

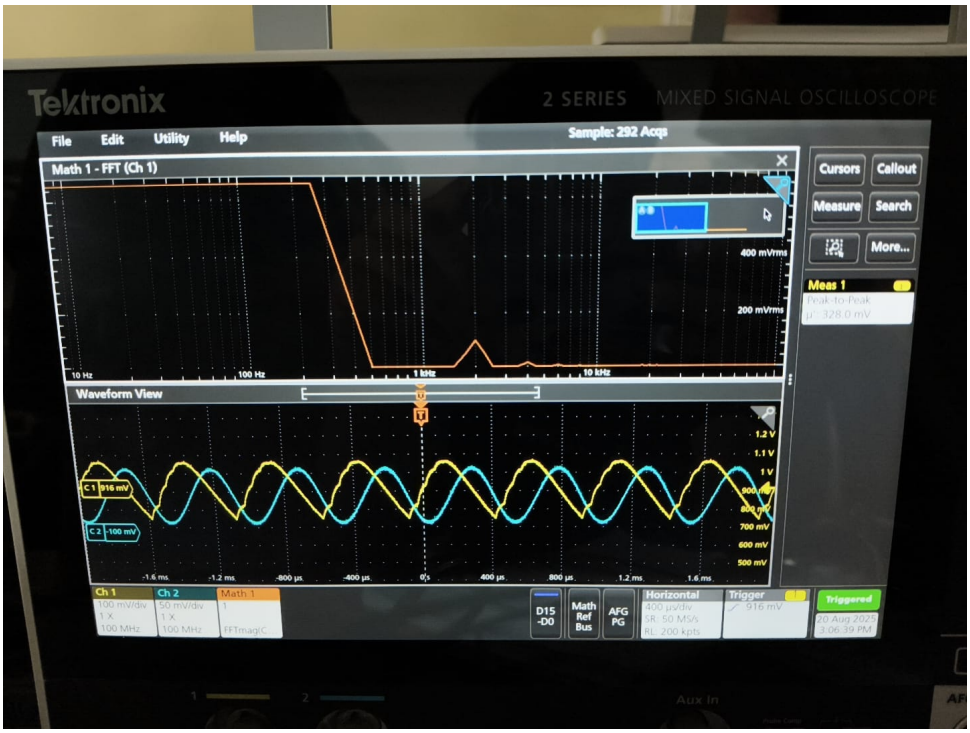


Figure 7: Input AM signal (Blue) and Recovered Message Signal (Yellow) with its FFT

Figure 7 shows the input AM wave (in blue) and the final recovered message signal from the low-pass filter output (in yellow). The recovered signal clearly matches the

envelope of the input wave, demonstrating successful demodulation, with some minor distortion.And its FFT showing peak at 5 kHz which corresponds to frequency of message signal.

5.2 Log-Antilog Modulator Performance

The message signal was set to 300 mV_{p-p} with a 300 mV DC offset at 2 kHz. The carrier signal was 300 mV_{p-p} with a 600 mV DC offset at 20 kHz.



Figure 8: Output of Log Circuit 1

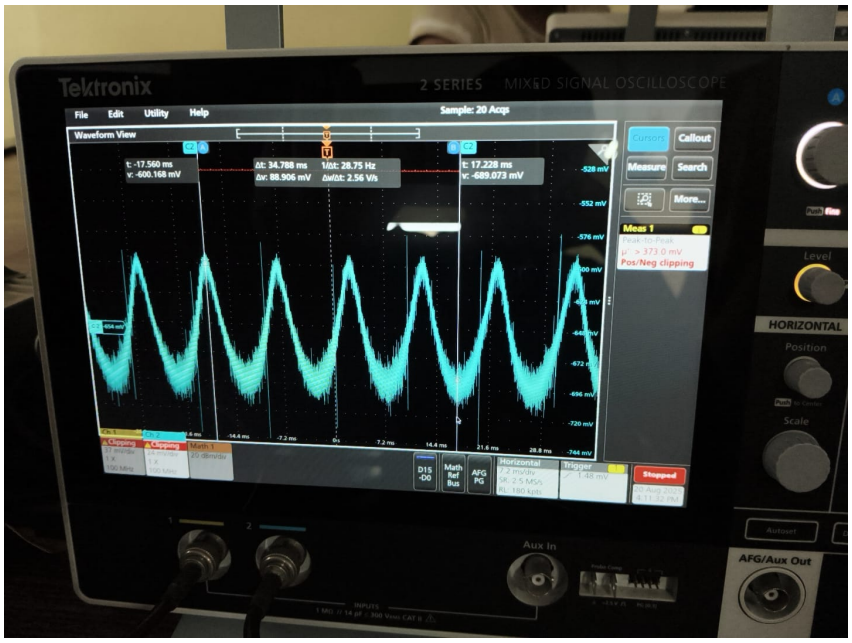


Figure 9: Output of Log Circuit 2

As seen in Figure 8 & 9, the Output of log circuit are not as expected ,lot of noise can be seen in signal ,which make it unsuitable for further processing.

6 Discussion

The experiment successfully demonstrated demodulation via an envelope detector but modulation via a log-antilog circuit was not succesfull due to some unknown reasons .

1. For the **log-antilog modulator**, providing correct DC offsets was critical. The offsets must be large enough to prevent the total signal from becoming negative (which would reverse-bias the diode), but not so large that they push the diode deep into its linear region, where the desired exponential I-V characteristic is lost.

2. The **envelope detector** is effective only for under-modulated signals ($m < 1$). If the signal is over-modulated ($m > 1$), phase reversal occurs at the zero-crossings, and the simple envelope detector cannot track this, leading to severe distortion in the demodulated output.

3. For proper envelope detection, the amplitude of the modulated signal must always remain above the diode's cut-in voltage ($\approx 0.7V$). An amplifier stage before the detector was used to boost the signal level to ensure this condition was met.

4. The output of the log circuit were having high noise which made it difficult to add both the signals and the output waveform was not as expected .Later we found out that it was because we connected the terminals in reverese polarity which caused the output to be negative . causing antilog circuit to give wrong output.

5. While the log-antilog method is conceptually elegant, it is power-intensive due to the required DC offsets. Furthermore, the anti-log stage is sensitive to DC levels at its input, which can drive the op-amp to saturation. This makes the circuit less practical for real-world applications without additional protective and level-shifting circuitry.

——END OF REPORT——